

Tokens

Store

The Store Screen, which can be accessed through the drawer on the home screen, allows users to purchase more tokens, with the intent of being able to charge users real money in exchange for tokens, although this element has not yet been implemented. The screen consists of some text at the top informing the user how many tokens they currently have in their account, and below that are a set of 6 buttons that the user can press to buy tokens, with the buttons adding 5, 10, 20, 50, 100, or 500 tokens to their account, in that order. The buttons also indicate how much they would add. When the user clicks on a button to add tokens, a modal appears to inform the user that they are about to purchase tokens. This was added to allow for a natural spot to exchange the tokens for real currency.

Usage

When implemented, the tokens can be used as a way to charge for use of certain features, or to buy recipes. For example, the parser in the Add Recipe screen costs 5 tokens to use (this value can be changed, but a multiple of 5 is recommended).

Implementation

Tokens are stored in Firebase, in the Firestore Database, under the User Perms collection, as a number named `tokens`. The tokens are accessed in the Flutter app through 3 functions in `lib/services/auth_services.dart`, named `getTokenCount`, `setTokenCount`, and `subtractTokenCount`, which are named according to what they do. The functions can be accessed anywhere by using `GetIt.I<AuthService>().functionName()`.

- `getTokenCount` takes in no arguments and returns a `Future<int?>`. This function does not throw any exceptions. This function returns null if the user is not logged in or does not exist in Firebase, otherwise, it returns the amount of tokens the user has, which should always be ≥ 0 .
- `setTokenCount` takes in an integer value named `numTokens`, which is used to set the number of tokens the user has. The function returns a `Future<void>`. This function does not throw any exceptions. This function does not add or subtract tokens to the user's account, this needs to be done before calling the function. This function does not check if the number passed to it is negative. The function will not change the token count if the user is not signed in or does not exist in Firebase.
- `subtractTokenAmount` takes in one integer argument which is the number of tokens to be removed from the user's account in Firebase. This function returns a `Future<void>`. This function does not check if the number passed to it is negative (this

could be used to add tokens, if so desired). The function will not change the token count if the user is not signed in or does not exist in Firebase.